

TransitOps

Complete Database Schema Reference

This is the consolidated schema for TransitOps, covering every entity introduced across the brief and the subsequent dashboard, financial, assignment-algorithm, and rating-score design work: 13 tables and 6 enumerated types, with field-level types, constraints, and descriptions, followed by an entity-relationship summary. Tables marked **core** are directly written to by users through the application; tables marked **derived** are computed/aggregated and typically refreshed by a batch job or recomputed on a triggering event rather than edited directly.

1. Enumerated Types

Shared value sets referenced by multiple tables below.

Enum	Values
RoleEnum	FLEET_MANAGER, DRIVER, SAFETY_OFFICER, FINANCIAL_ANALYST
VehicleStatus	AVAILABLE, ON_TRIP, IN_SHOP, RETIRED
DriverStatus	AVAILABLE, ON_TRIP, OFF_DUTY, SUSPENDED
TripStatus	DRAFT, DISPATCHED, COMPLETED, CANCELLED
ExpenseCategory	TOLL, FINE, PARKING, OTHER
IncidentSeverity	LOW, MEDIUM, HIGH

2. Core Tables

Written to directly by users through the application UI.

2.1 User

Authentication and RBAC. One row per login account.

Field	Type	Constraints	Description
id	UUID	PK	Primary key
name	string	NOT NULL	Full name
email	string	UNIQUE, NOT NULL	Login identifier
passwordHash	string	NOT NULL	Hashed password, never stored in plaintext
role	RoleEnum	NOT NULL	Drives RBAC route/permission checks
isActive	boolean	DEFAULT true	Deactivated users cannot log in
createdAt	datetime	DEFAULT now()	Account creation timestamp

2.2 Vehicle

Master registry of fleet assets.

Field	Type	Constraints	Description
id	UUID	PK	Primary key
registrationNumber	string	UNIQUE, NOT NULL	Plate/registration number
modelName	string	NOT NULL	Vehicle name/model
type	string	NOT NULL	Vehicle category (van, truck, etc.)
maxLoadCapacity	float	NOT NULL, > 0	Max cargo weight, kg
odometer	float	DEFAULT 0	Current mileage
acquisitionCost	float	NOT NULL	Purchase cost; feeds ROI calc
acquisitionDate	date	NOT NULL	Feeds average vehicle age KPI
status	VehicleStatus	DEFAULT AVAILABLE	Drives dispatch-pool visibility
lastServiceOdometer	float	DEFAULT 0	Set automatically when maintenance closes
serviceIntervalKm	float	NOT NULL	Manually configured; drives maintenance prediction

2.3 Driver

Driver profiles and compliance data.

Field	Type	Constraints	Description
id	UUID	PK	Primary key
userId	UUID	FK → User.id, NULLABLE	Link to login account, if driver has app access

Field	Type	Constraints	Description
name	string	NOT NULL	Full name
licenseNumber	string	UNIQUE, NOT NULL	Government license ID
licenseCategory	string	NOT NULL	Determines vehicle-type eligibility
licenseExpiryDate	date	NOT NULL	Blocks dispatch when past today
contactNumber	string	NOT NULL	Phone contact
safetyScore	float	DEFAULT 100	0–100; manual or telematics-set
status	DriverStatus	DEFAULT AVAILABLE	Drives dispatch-pool visibility

2.4 Trip

Central operational record; one row per delivery job.

Field	Type	Constraints	Description
id	UUID	PK	Primary key
vehicleId	UUID	FK → Vehicle.id, NOT NULL	Assigned vehicle
driverId	UUID	FK → Driver.id, NOT NULL	Assigned driver
createdByUserId	UUID	FK → User.id, NOT NULL	Who created the trip
source	string	NOT NULL	Origin location
destination	string	NOT NULL	Destination location
cargoWeight	float	NOT NULL, ≤ Vehicle.maxLoadCapacity	Validated at creation
plannedDistance	float	NOT NULL	Estimated distance, km
actualDistance	float	NULLABLE	Set on completion
plannedDuration	float	NULLABLE	Hours; feeds on-time rate KPI
actualDuration	float	NULLABLE	Derived: completedAt – dispatchedAt
fuelConsumed	float	NULLABLE	Set on completion
revenue	float	NULLABLE	Manually entered; feeds profit/ROI KPIs
status	TripStatus	DEFAULT DRAFT	Draft → Dispatched → Completed/Cancelled
dispatchedAt	datetime	NULLABLE	Set when status becomes Dispatched
completedAt	datetime	NULLABLE	Set when status becomes Completed
createdAt	datetime	DEFAULT now()	Record creation timestamp

2.5 MaintenanceLog

Service and repair history per vehicle.

Field	Type	Constraints	Description
id	UUID	PK	Primary key
vehicleId	UUID	FK → Vehicle.id, NOT NULL	Vehicle serviced
serviceType	string	NOT NULL	e.g. Oil Change, Tire Rotation
description	string	NULLABLE	Free-text notes
cost	float	NOT NULL	Feeds operational cost KPI
serviceDate	date	NOT NULL	Date service performed
expectedCompletionDate	date	NULLABLE	Manually entered ETA while active
isPlanned	boolean	DEFAULT true	Scheduled service vs. unplanned repair
isActive	boolean	DEFAULT true	true while vehicle is In Shop; false once closed

2.6 FuelLog

Fuel purchases per vehicle, optionally tied to a trip.

Field	Type	Constraints	Description
id	UUID	PK	Primary key
vehicleId	UUID	FK → Vehicle.id, NOT NULL	Vehicle fueled
tripId	UUID	FK → Trip.id, NULLABLE	Trip during which fuel was logged
liters	float	NOT NULL, > 0	Volume purchased
cost	float	NOT NULL	Feeds operational cost KPI
date	date	NOT NULL	Purchase date

2.7 Expense

Non-fuel operational costs (tolls, fines, parking, other).

Field	Type	Constraints	Description
id	UUID	PK	Primary key
vehicleId	UUID	FK → Vehicle.id, NOT NULL	Vehicle the expense applies to
tripId	UUID	FK → Trip.id, NULLABLE	Trip the expense applies to, if specific
category	ExpenseCategory	NOT NULL	TOLL / FINE / PARKING / OTHER
amount	float	NOT NULL	Feeds operational cost KPI

Field	Type	Constraints	Description
date	date	NOT NULL	Expense date
notes	string	NULLABLE	Free-text notes

2.8 VehicleDocument

Insurance, registration, and permit expiry tracking.

Field	Type	Constraints	Description
id	UUID	PK	Primary key
vehicleId	UUID	FK → Vehicle.id, NOT NULL	Vehicle the document belongs to
documentType	string	NOT NULL	Insurance, Registration, Permit, etc.
documentNumber	string	NULLABLE	Reference number
expiryDate	date	NOT NULL	Drives expiry-tracker widget
fileUrl	string	NULLABLE	Link to scanned document, if uploaded

2.9 Incident

Safety incidents/violations logged against a driver.

Field	Type	Constraints	Description
id	UUID	PK	Primary key
driverId	UUID	FK → Driver.id, NOT NULL	Driver involved
tripId	UUID	FK → Trip.id, NULLABLE	Trip during which it occurred, if applicable
severity	IncidentSeverity	NOT NULL	LOW / MEDIUM / HIGH
description	string	NOT NULL	What happened
status	string	DEFAULT OPEN	OPEN / RESOLVED
loggedByUserId	UUID	FK → User.id, NOT NULL	Safety Officer who logged it
occurredAt	datetime	NOT NULL	When the incident happened

3. Derived Tables

Computed from the core tables above via aggregation. Refreshed by a nightly batch job or recomputed on a triggering event (trip completed, maintenance closed, incident logged) rather than edited directly by users.

3.1 VehicleFinancialSummary

Per-vehicle profit/cost rollup backing the Financial Analyst dashboard.

Field	Type	Constraints	Description
vehicleId	UUID	PK, FK → Vehicle.id	One row per vehicle
totalRevenue	float	DERIVED	SUM(Trip.revenue)
totalFuelCost	float	DERIVED	SUM(FuelLog.cost)
totalMaintenanceCost	float	DERIVED	SUM(MaintenanceLog.cost)
totalOtherExpenses	float	DERIVED	SUM(Expense.amount)
totalOperationalCost	float	DERIVED	fuel + maintenance + other
netProfit	float	DERIVED	totalRevenue – totalOperationalCost
roi	float	DERIVED	netProfit / Vehicle.acquisitionCost
fuelEfficiency	float	DERIVED	SUM(Trip.actualDistance) / SUM(FuelLog.liters)
computedAt	datetime	DEFAULT now()	Last recompute timestamp

3.2 DriverRatingScore

Consolidated 1–3 star driver rating (see Rating Scores document for formulas).

Field	Type	Constraints	Description
driverId	UUID	PK, FK → Driver.id	One row per driver
safetyComponent	float	DERIVED, 0–1	Driver.safetyScore / 100
complianceComponent	float	DERIVED, 0–1	License validity window check
completionComponent	float	DERIVED, 0–1	Completed / (Completed + Cancelled)
onTimeComponent	float	DERIVED, 0–1	On-time completions / total completions
incidentFreeComponent	float	DERIVED, 0–1	Rolling 90-day incident penalty
weightedScore	float	DERIVED, 0–1	Weighted sum of the five components above
starRating	float	DERIVED, 1.0–3.0	Mapped from weightedScore, 0.5 steps
computedAt	datetime	DEFAULT now()	Last recompute timestamp

3.3 VehicleRatingScore

Consolidated 1–3 star vehicle rating (see Rating Scores document for formulas).

Field	Type	Constraints	Description
vehicleId	UUID	PK, FK → Vehicle.id	One row per vehicle
maintenanceComponent	float	DERIVED, 0–1	Based on odometer vs. service interval

Field	Type	Constraints	Description
utilizationComponent	float	DERIVED, 0–1	Trips vs. fleet average, last 30 days
fuelEfficiencyComponent	float	DERIVED, 0–1	vs. fleet average distance/liter
breakdownComponent	float	DERIVED, 0–1	Based on unplanned maintenance frequency
roiComponent	float	DERIVED, 0–1	Normalized ROI across fleet, min-max scaled
weightedScore	float	DERIVED, 0–1	Weighted sum of the five components above
starRating	float	DERIVED, 1.0–3.0	Mapped from weightedScore, 0.5 steps
computedAt	datetime	DEFAULT now()	Last recompute timestamp

4. Entity-Relationship Summary

User	1	----	*	Trip	(createdByUserId)
User	1	----	0..1	Driver	(optional link to login account)
User	1	----	*	Incident	(loggedByUserId)
Vehicle	1	----	*	Trip	(vehicleId)
Vehicle	1	----	*	MaintenanceLog	(vehicleId)
Vehicle	1	----	*	FuelLog	(vehicleId)
Vehicle	1	----	*	Expense	(vehicleId)
Vehicle	1	----	*	VehicleDocument	(vehicleId)
Vehicle	1	----	1	VehicleFinancialSummary	
Vehicle	1	----	1	VehicleRatingScore	
Driver	1	----	*	Trip	(driverId)
Driver	1	----	*	Incident	(driverId)
Driver	1	----	1	DriverRatingScore	
Trip	1	----	0..*	FuelLog	(tripId, optional)
Trip	1	----	0..*	Expense	(tripId, optional)
Trip	1	----	0..*	Incident	(tripId, optional)

4.1 Cross-cutting business rules enforced at the application layer

- Vehicle.registrationNumber and Driver.licenseNumber are globally unique.
- Trip.cargoWeight must not exceed the assigned Vehicle.maxLoadCapacity at creation time.
- Only Vehicles with status = AVAILABLE and Drivers with status = AVAILABLE and licenseExpiryDate in the future are selectable for dispatch.
- Dispatching a Trip sets both Vehicle.status and Driver.status to ON_TRIP; completing or cancelling a dispatched Trip reverts both to AVAILABLE.
- Creating an active MaintenanceLog (isActive = true) sets Vehicle.status to IN_SHOP; closing it (isActive = false) reverts to AVAILABLE unless the vehicle is RETIRED.
- VehicleFinancialSummary, DriverRatingScore, and VehicleRatingScore are never written to directly by the application UI — only by the batch recompute job.